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COHORT A

```
MariaDB [(none)]> CREATE TABLE Department(deptNo INT PRIMARY KEY,  
deptName VARCHAR(100) NOT NULL, mgrEmpNo INT);  
ERROR 1046 (3D000): No database selected  
MariaDB [(none)]> Create database projectSchema;  
Query OK, 1 row affected (0.009 sec)
```

```
MariaDB [(none)]> Use projectSchema;  
Database changed  
MariaDB [projectSchema]> Create Table Department(deptNo INT PRIMARY  
KEY, deptName VARCHAR(100) NOT NULL, mgrEmpno INT);  
Query OK, 0 rows affected (0.044 sec)
```

```
MariaDB [projectSchema]> Create Table Employee (empNo INT PRIMARY  
KEY, fName VARCHAR(50) NOT NULL, lName VARCHAR(50) NOT NULL, address  
VARCHAR(150), DOB DATE, sex CHAR(1) NOT NULL CHECK (sex IN ('M',  
'F')), position VARCHAR(50) CHECK (position IN ('Manager', 'Team  
Leader', 'Analyst', 'Software Developer')), deptNo INT, FOREIGN KEY  
(deptNo) REFERENCES Department(deptNo));  
Query OK, 0 rows affected (0.031 sec)
```

```
MariaDB [projectSchema]> ALTER TABLE Department  
-> ADD CONSTRAINT fk_dept_manager  
-> FOREIGN KEY (mgrEmpNo) REFERENCES Employee(empNo);  
Query OK, 0 rows affected (0.056 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [projectSchema]> Create Table Project( projNO INT PRIMARY  
KEY, projName VARCHAR(100) NOT NULL, deptNo INT, FOREIGN KEY  
(deptNo) REFERENCES Department(deptNo));  
Query OK, 0 rows affected (0.029 sec)
```

```
MariaDB [projectSchema]> Create Table WorksOn (empNo INT, projNo  
INT, dateWorked DATE, hoursWorked INT CHECK (hoursWorked >= 0 AND  
hoursWorked <= 40), PRIMARY KEY (empNo, projNo, dateWorked), FOREIGN  
KEY (empNo) REFERENCES Employee(empNo), FOREIGN KEY (projNo)  
REFERENCES Project(projNo));  
Query OK, 0 rows affected (0.029 sec)
```

```
MariaDB [projectSchema]> CREATE VIEW FemaleManagerProjects AS SELECT  
-> p.projNO, p.projName, p.deptNo  
-> FROM Project p  
-> Join Department d ON p.deptNo = d.deptNo  
-> JOIN Employee e ON d.mgrEmpNo = e.empNo  
-> WHERE e.sex = 'F'  
-> ORDER BY p.projNO;  
Query OK, 0 rows affected (0.032 sec)
```

```
MariaDB [projectSchema]> INSERT INTO Department (deptNO, deptName)  
VALUES  
-> (1, 'Engineering'),
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    -> (2, 'Finance'),
    -> (3, 'Marketing');
Query OK, 3 rows affected (0.005 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
MariaDB [projectSchema]> INSERT INTO Employee VALUES
    -> (1, 'Ama', 'Owusu', 'Accra', '1990-05-01', 'F', 'Manager',
1),
    -> (2, 'Kofi', 'Mensah', 'Kumasi', '1985-03-15', 'M', 'Team
Leader', 1),
    -> (3, 'Abena', 'Asante', 'Takoradi', '1992-07-20', 'F',
'Analyst', 2),
    -> (4, 'Kwame', 'Boateng', 'Accra', '1988-11-30', 'M', 'Software
Developer', 2),
    -> (5, 'Yaa', 'Darko', 'Cape Coast', '1995-01-10', 'F',
'Manager', 3);
Query OK, 5 rows affected (0.002 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
MariaDB [projectSchema]> UPDATE Department SET mgrEmpNo = 1 WHERE
deptNo = 1;
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [projectSchema]> UPDATE Department SET mgrEmpNo = 3 WHERE
deptNo = 2;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [projectSchema]> UPDATE Department SET mgrEmpNo = 5 WHERE
deptNo = 3;
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [projectSchema]> INSERT INTO Project VALUES
    -> (101, 'SystemX', 1),
    -> (102, 'BudgetTool', 2),
    -> (103, 'AdCampaign', 3),
    -> (104, 'NetworkUpgrade', 1);
Query OK, 4 rows affected (0.001 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
MariaDB [projectSchema]> INSERT INTO WorksOn VALUES
    -> (1, 101, '2025-01-10', 8),
    -> (1, 101, '2025-01-11', 7),
    -> (2, 101, '2025-01-10', 6),
    -> (2, 104, '2025-01-12', 8),
    -> (3, 102, '2025-01-10', 5),
    -> (3, 102, '2025-01-11', 8),
    -> (4, 102, '2025-01-10', 7),
    -> (5, 103, '2025-01-10', 6),
    -> (1, 104, '2025-01-13', 8);
Query OK, 9 rows affected (0.001 sec)
Records: 9  Duplicates: 0  Warnings: 0
```

MariaDB [projectSchema]> INSERT INTO Part VALUES

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-> (1, 'ContractA', 1500.00),
-> (1, 'ContractB', 800.00),
-> (2, 'ContractA', 1200.00),
-> (3, 'ContractC', 500.00),
-> (4, 'ContractA', 2000.00),
-> (4, 'ContractB', 1100.00);
```

Query OK, 6 rows affected (0.001 sec)

Records: 6 Duplicates: 0 Warnings: 0

MariaDB [projectSchema]> SELECT * FROM FemaleManagerProjects;

projNo	projName	deptNo
101	SystemX	1
102	BudgetTool	2
103	AdCampaign	3
104	NetworkUpgrade	1

4 rows in set (0.001 sec)

MariaDB [projectSchema]> SELECT * FROM FemaleManagerProjects;

projNo	projName	deptNo
101	SystemX	1
104	NetworkUpgrade	1
102	BudgetTool	2
103	AdCampaign	3

4 rows in set (0.003 sec)

MariaDB [projectSchema]> CREATE VIEW EmpProjectDetails AS SELECT

```
-> e.empNo,
-> e.fName,
-> e.lName,
-> p.projName,
-> w.hoursWorked
-> FROM Employee e
-> JOIN WorksOn w ON e.empNo = w.empNo
-> JOIN Project p ON w.projNo = p.projNo;
```

Query OK, 0 rows affected (0.023 sec)

MariaDB [projectSchema]> SELECT * FROM EmpProjectDetails;

empNo	fName	lName	projName	hoursWorked
1	Ama	Owusu	SystemX	8
1	Ama	Owusu	SystemX	7
1	Ama	Owusu	NetworkUpgrade	8
2	Kofi	Mensah	SystemX	6
2	Kofi	Mensah	NetworkUpgrade	8
3	Abena	Asante	BudgetTool	5

3	Abena	Asante	BudgetTool	8
4	Kwame	Boateng	BudgetTool	7
5	Yaa	Darko	AdCampaign	6

9 rows in set (0.001 sec)

MariaDB [projectSchema]> CREATE VIEW EmpProject(empNo, projNo, totalHours)

-> AS SELECT w.empNo, w.projNo, SUM(hoursWorked)

-> FROM Employee e, Project p, WorksOn w

-> WHERE e.empNo = w.empNo AND p.projNo = w.projNo

-> GROUP BY w.empNo, w.projNo;

Query OK, 0 rows affected (0.024 sec)

MariaDB [projectSchema]> SELECT * FROM EmpProject;

empNo	projNo	totalHours
1	101	15
1	104	8
2	101	6
2	104	8
3	102	13
4	102	7
5	103	6

7 rows in set (0.001 sec)

MariaDB [projectSchema]> SELECT projNo FROM EmpProject WHERE projNo = 101;

projNo
101
101

2 rows in set (0.002 sec)

MariaDB [projectSchema]> SELECT COUNT(projNo) FROM EmpProject WHERE empNo = 1;

COUNT(projNo)
2

1 row in set (0.001 sec)

MariaDB [projectSchema]> SELECT empNo, totalHours

-> FROM EmpProject

-> GROUP BY empNo;

Empty set (0.001 sec)

MariaDB [projectSchema]> SELECT empNo, SUM(totalHours) AS grandTotalHours

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-> FROM EmpProject
-> GROUP BY empNo;

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empNo	grandTotalHours
1	23
2	14
3	13
4	7
5	6

5 rows in set (0.002 sec)

```

MariaDB [projectSchema]> CREATE TABLE Part (partNo INT, contract
VARCHAR(50), partCost DECIMAL(10,2), PRIMARY KEY (partNo,
contract));
Query OK, 0 rows affected (0.043 sec)

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MariaDB [projectSchema]> CREATE VIEW ExpensiveParts (partNo)
-> AS SELECT DISTINCT partNo
-> FROM Part
-> WHERE partCost > 1000;
Query OK, 0 rows affected (0.020 sec)

```

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SELECT * FROM ExpensiveParts;

```

partNo
1
2
4

3 rows in set (0.001 sec)

```

MariaDB [projectSchema]> SELECT 'Department' AS 'Table', COUNT(*) AS
'Row Count' FROM Department
-> UNION
-> SELECT 'Employee', COUNT(*) FROM Employee
-> UNION
-> SELECT 'Project', COUNT(*) FROM Project
-> UNION
-> SELECT 'WorksOn', COUNT(*) FROM WorksOn
-> UNION
-> SELECT 'Part', COUNT(*) FROM Part
-> UNION
-> SELECT 'FemaleManagerProjects', COUNT(*) FROM
FemaleManagerProjects
-> UNION
-> SELECT 'EmpProject', COUNT(*) FROM EmpProject
-> UNION
-> SELECT 'EmpProjectDetails', COUNT(*) FROM EmpProjectDetails;

```

Table	Row Count
Department	1
Employee	1
Project	1
WorksOn	1
Part	1
FemaleManagerProjects	1
EmpProject	1
EmpProjectDetails	1

Department	3	
Employee	5	
Project	4	
WorksOn	9	
Part	6	
FemaleManagerProjects	4	
EmpProject	7	
EmpProjectDetails	9	
+-----+		
8 rows in set (0.008 sec)		